Chapter 7 – OpenSSH Server & Remote Administration

Objective

Managing a server directly through the VirtualBox console can become inconvenient, especially when working with multiple virtual machines. OpenSSH allows administrators to securely manage Ubuntu Server remotely from another computer using the Secure Shell (SSH) protocol.

In this chapter, you will learn how to:

- Understand SSH.
- Install and verify OpenSSH Server.
- Manage the SSH service.
- Connect from Windows using Windows Terminal.
- Securely transfer files.
- Troubleshoot common SSH problems.

7.1 What is SSH?

SSH (Secure Shell) is a secure network protocol used to remotely access and manage Linux servers over a network.

Instead of sitting in front of the server, an administrator can securely connect from another computer and execute commands as if working directly on the machine.

SSH encrypts all communication between the client and the server, protecting usernames, passwords, commands, and transferred data from interception.

7.2 Why Use SSH?

SSH provides several advantages:

- Secure remote administration
- Encrypted communication

- Easy file transfer using SCP or SFTP
- Supports automation through scripts
- Widely used in enterprise environments

For cybersecurity professionals, SSH is the standard method of managing Linux servers.

7.3 Verify OpenSSH Installation

Ubuntu Server usually installs OpenSSH if the option was selected during installation.

Verify whether the SSH server package is installed.

dpkg -l | grep openssh-server

Expected output:

ii openssh-server

If nothing is returned, install it using:

sudo apt update

sudo apt install openssh-server -y

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:49:9d:94 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 86391sec preferred_lft 86391sec
    inet6 fd17:625c:f037:2:a00:27ff:fe49:9d94/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86393sec preferred_lft 14393sec
    inet6 fe80::a00:27ff:fe49:9d94/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:30:bd:69 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
       valid_lft 591sec preferred_lft 591sec
    inet6 fe80::a00:27ff:fe30:bd69/64 scope link
       valid_lft forever preferred_lft forever
kashif@ubunv22:~$ dpkg -l | grep ssh
ii  libssh-4:amd64                         0.9.6-2ubuntu0.22.04.7                                 amd64
     tiny C SSH library (OpenSSL flavor)
ii  openssh-client                         1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) client, for secure access to remote machines
ii  openssh-server                         1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) server, for secure access from remote machines
ii  openssh-sftp-server                    1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) sftp server module, for SFTP access from remote machines
ii  ssh-import-id                          5.11-0ubuntu1                                          all
     securely retrieve an SSH public key and install it locally
kashif@ubunv22:~$ dpkg -l | grep openssh-server
ii  openssh-server                         1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) server, for secure access from remote machines
kashif@ubunv22:~$
```

7.4 Verify the SSH Service

Check the current status of the SSH service.

sudo systemctl status ssh

Expected output:

Active: active (running)

The **active (running)** status confirms that the SSH server is ready to accept remote connections.

```
    inet6 fe80::a00:27ff:fe30:bd69/64 scope link
       valid_lft forever preferred_lft forever
kashif@ubunv22:~$ dpkg -l | grep ssh
ii  libssh-4:amd64                        0.9.6-2ubuntu0.22.04.7                                 amd64
     tiny C SSH library (OpenSSL flavor)
ii  openssh-client                        1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) client, for secure access to remote machines
ii  openssh-server                        1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) server, for secure access from remote machines
ii  openssh-sftp-server                   1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) sftp server module, for SFTP access from remote machines
ii  ssh-import-id                         5.11-0ubuntu1                                          all
     securely retrieve an SSH public key and install it locally
kashif@ubunv22:~$ dpkg -l | grep openssh-server
ii  openssh-server                        1:8.9p1-3ubuntu0.16                                    amd64
     secure shell (SSH) server, for secure access from remote machines
kashif@ubunv22:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-07-16 06:17:45 UTC; 23min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 712 (sshd)
      Tasks: 1 (limit: 5776)
     Memory: 6.8M
        CPU: 1.033s
     CGroup: /system.slice/ssh.service
             └─712 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 06:17:41 ubunv22 systemd[1]: Starting OpenBSD Secure Shell server...
Jul 16 06:17:45 ubunv22 sshd[712]: Server listening on 0.0.0.0 port 22.
Jul 16 06:17:45 ubunv22 sshd[712]: Server listening on :: port 22.
Jul 16 06:17:45 ubunv22 systemd[1]: Started OpenBSD Secure Shell server.
Jul 16 06:36:17 ubunv22 sshd[1235]: Accepted password for kashif from 192.168.56.1 port 55160 ssh2
Jul 16 06:36:17 ubunv22 sshd[1235]: pam_unix(sshd:session): session opened for user kashif(uid=1000>
lines 1-18/18 (END)
kashif@ubunv22:~$ _
```

7.5 Enable SSH at Startup

Ensure that the SSH service starts automatically when Ubuntu boots.

sudo systemctl enable ssh

Verify:

sudo systemctl is-enabled ssh

Expected output:

enabled

7.6 Determine the Server IP Address

Display all IP addresses assigned to the server.

ip addr

Example:

enp0s3

10.0.2.15

enp0s8

192.168.56.10

The **Host-Only IP** (192.168.56.10) will be used for remote administration within the home lab.

7.7 Connect from Windows

Open **Windows Terminal** or **PowerShell**.

Use the following command:

ssh kashifkhan@192.168.56.10

On the first connection, SSH displays the server fingerprint.

Are you sure you want to continue connecting (yes/no)?

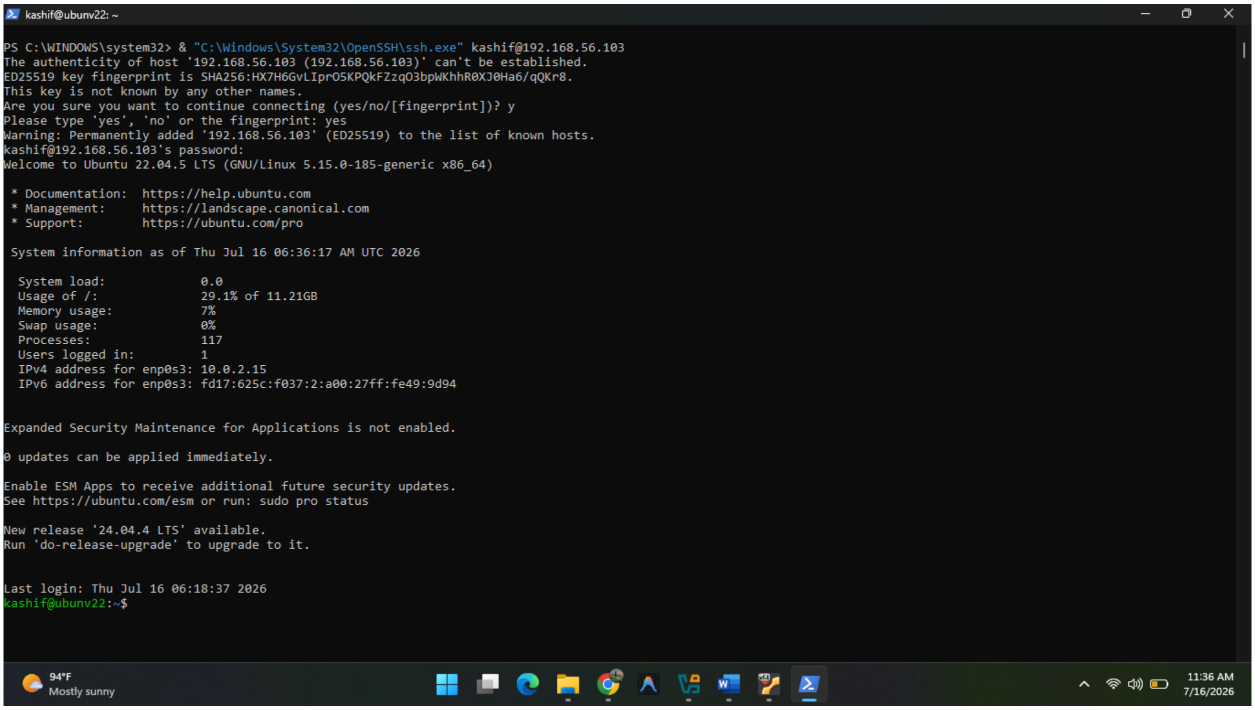
Type:

yes

Then enter the user password.

If successful, the prompt changes to:

kashifkhan@ubunv22:~$



7.8 Copy Files Securely

SSH also allows secure file transfers using **SCP**.

Example:

scp report.txt kashifubunv22@192.168.56.10:/home/kashifubunv22/

This copies **report.txt** from Windows to Ubuntu Server.

7.9 Useful SSH Commands

Command	**Description**
ssh user@ip	Connect to a remote server
scp file user@ip:path	Copy a file to the server
exit	Disconnect from the SSH session
hostname	Display the server hostname

whoami	Display the current logged-in user

7.10 Common SSH Problems

Problem	Possible Cause	Solution
Connection refused	SSH service is stopped	sudo systemctl start ssh
Permission denied	Incorrect username or password	Verify credentials
Host unreachable	Network configuration issue	Check Host-Only adapter and IP address
Command not found: ssh	OpenSSH Client missing on Windows	Install OpenSSH Client from Windows Optional Features

---

Best Practices

- Use strong passwords.
- Keep OpenSSH updated.
- Disable password authentication when using SSH keys in production.
- Use Host-Only networking for local lab access.
- Verify the SSH service after every reboot.

Chapter Summary

In this chapter, you learned how to:

- Understand SSH.
- Verify and install OpenSSH Server.
- Manage the SSH service.

- Connect from Windows.
- Transfer files using SCP.
- Troubleshoot common SSH issues.

One-Line Summary

OpenSSH enables secure remote administration of Ubuntu Server, allowing administrators to manage systems efficiently without requiring direct access to the virtual machine console.